

Math 158 Project Proposal

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Introduction

```
library(readr)
sleep_data <- read_csv("Sleep_health_and_lifestyle_dataset.csv")
sleep_data
```

```
# A tibble: 374 x 13
  `Person ID` Gender   Age Occupation      `Sleep Duration` `Quality of Sleep`
      <dbl> <chr>   <dbl> <chr>          <dbl>          <dbl>
1           1 Male     27 Software Engine~      6.1             6
2           2 Male     28 Doctor              6.2             6
3           3 Male     28 Doctor              6.2             6
4           4 Male     28 Sales Represent~    5.9             4
5           5 Male     28 Sales Represent~    5.9             4
6           6 Male     28 Software Engine~    5.9             4
7           7 Male     29 Teacher              6.3             6
8           8 Male     29 Doctor              7.8             7
9           9 Male     29 Doctor              7.8             7
10          10 Male     29 Doctor              7.8             7
# i 364 more rows
# i 7 more variables: `Physical Activity Level` <dbl>, `Stress Level` <dbl>,
#   `BMI Category` <chr>, `Blood Pressure` <chr>, `Heart Rate` <dbl>,
#   `Daily Steps` <dbl>, `Sleep Disorder` <chr>
```

Sleep is an essential part of life. It allows the body and brain to rest and recover from daily activities. Sleep is meant to be as simple as closing your eyes, however, it is more complex when evaluating it through the lens of science. Sleeping well has positive outcomes for individuals who get between seven to nine hours of sleep. While this amount of sleep is recommended, it does not apply to everyone. Some people need less, and some people need more depending on the circumstances. However, sleeping less or not enough can lead to serious consequences

generally. The consequences can manifest as sleeping disorders, compromising the ability of the body and brain to function properly. Sleep disorders are common, and they affect the health and well-being of those who suffer from one.

According to Cleveland Clinic, sleep disorders refer to conditions which affect the quality, amount, and timing of a person's sleep. They also state that over 50 million Americans have them. Therefore, it is important to investigate why so many Americans are affected by this issue. We can do this using the sleep dataset obtained from Kaggle. Our sleep dataset contains the following variables: gender, age, occupation, sleep duration, quality of sleep, physical activity level, stress level, BMI (body mass index) category, blood pressure, heart rate, daily steps, and sleep disorder. In addition, there are a total of 374 individuals who were part of this study. Our project's central research questions are, "Is there an association between a person's background information and risk for developing sleep disorders?" While this is our primary research goal, for our project draft, we will be investigating the relationship between a person's resting heart rate and their background information until we have the statistical tools to answer the primary question. We suspect there is a relationship though given that existing medical literature has shown that BMI, physical activity levels, and stress levels significantly affect an individual's quality of sleep and heart rate.

Methods

For our project, we will be utilizing the sleep dataset. Given that this dataset includes background information for each individual, we are interested in predicting sleep disorder status given various predictor variables such as stress level, physical activity, sleep duration, age, BMI category, etc. This, however, would require a multinomial regression model given that our response variable, sleep disorder, is categorical. This will be our extension instead. For now though, we will be doing a simple regression analysis of the relationship between heart rate and an individual's background. To go about this goal, we will first start off with exploratory data analysis. This can take the forms of frequency tables, distribution plots, scatterplots, and correlation matrices. For a sparse linear model, we anticipate comparing its performance to the MLR model using scoring methods such as RMSE and cross-validation error. Lastly, exploring nonlinear relationships for the prediction of sleep quality given an individual's background is essential because it may be the case that physical activity improves sleep quality initially, but it dips after a certain threshold is achieved. This will be done by smoothing.

Our hypothesis for this project is that insomnia and sleep apnea will have different predictor variables with greater influence. For instance, we hypothesized that insomnia is linked more so to stress level, sleep duration, and quality of sleep due to these factors going hand in hand with mental health. According to the sleep foundation, 50% of people with insomnia also report having a mental health condition. As for sleep apnea, existing medical literature states that it has a greater correlation with physical health.

Extension

For our extension of a method not covered during class, we will be developing a multinomial logistic regression model given that we are interested in predicting sleep disorder status. Our hypothesis for this extension is that insomnia and sleep apnea will have different predictor variables with greater influence. For instance, we hypothesize that insomnia is linked more so to stress level, sleep duration, and quality of sleep due to these factors going hand in hand with mental health.. As for sleep apnea, existing medical literature states that it has a greater correlation with physical health. Therefore, perhaps it is more linked to bmi, daily steps, and resting blood pressure.

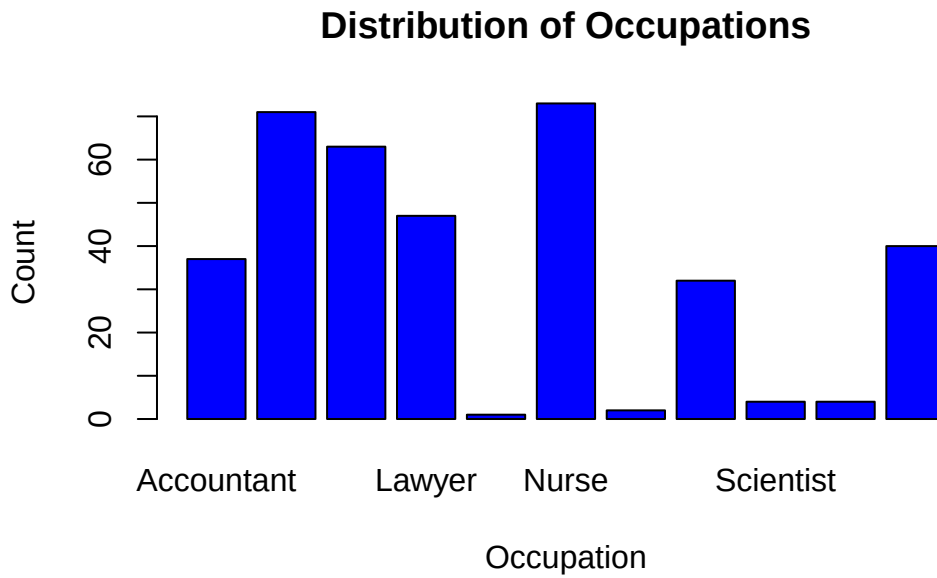
Results

Occupation Distribution among Study Participants

```
x <- sleep_data$Occupation
occupation_counts <- table(sleep_data$Occupation)
occupation_counts
```

Accountant	Doctor	Engineer
37	71	63
Lawyer	Manager	Nurse
47	1	73
Sales Representative	Salesperson	Scientist
2	32	4
Software Engineer	Teacher	
4	40	

```
barplot(occupation_counts,
        main = "Distribution of Occupations",
        xlab = "Occupation",
        ylab = "Count",
        col = "blue")
```



The distribution above shows a wide range of occupations, however, it is not balanced. Doctors and nurses are the highest participants with 71 and 73, respectively. Engineers are also greater compared to sales representatives. This means that the data is not distributed evenly, and that can lead to variations in the results.

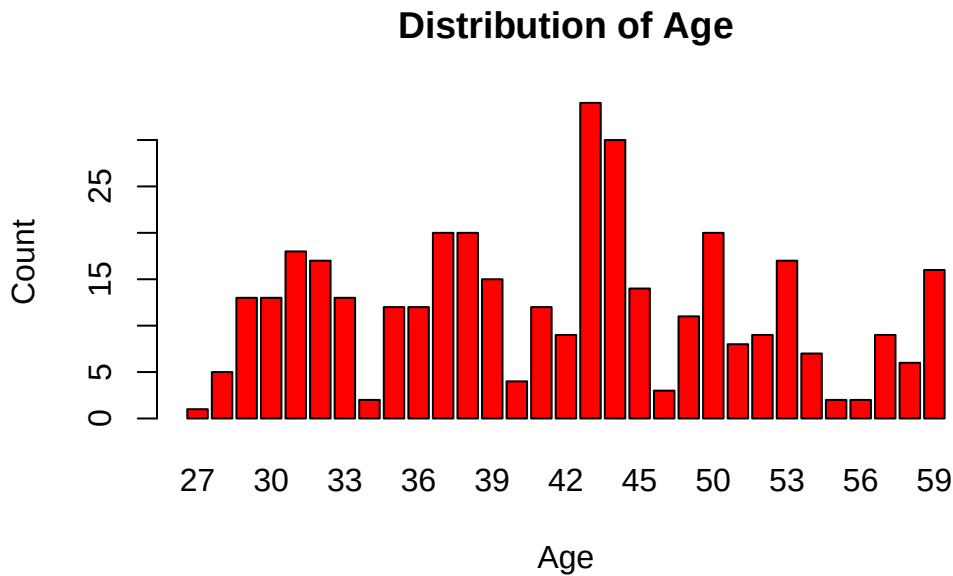
Age Distribution of Study Participants

```
y <- sleep_data$Age
age_counts <- table(sleep_data$Age)
age_counts
```

```
27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 48 49 50 51 52 53 54
1  5 13 13 18 17 13  2 12 12 20 20 15  4 12  9 34 30 14  3 11 20  8  9 17  7
55 56 57 58 59
2  2  9  6 16
```

```
barplot(age_counts,
        main = "Distribution of Age",
        xlab = "Age",
```

```
ylab = "Count",
col = "red")
```



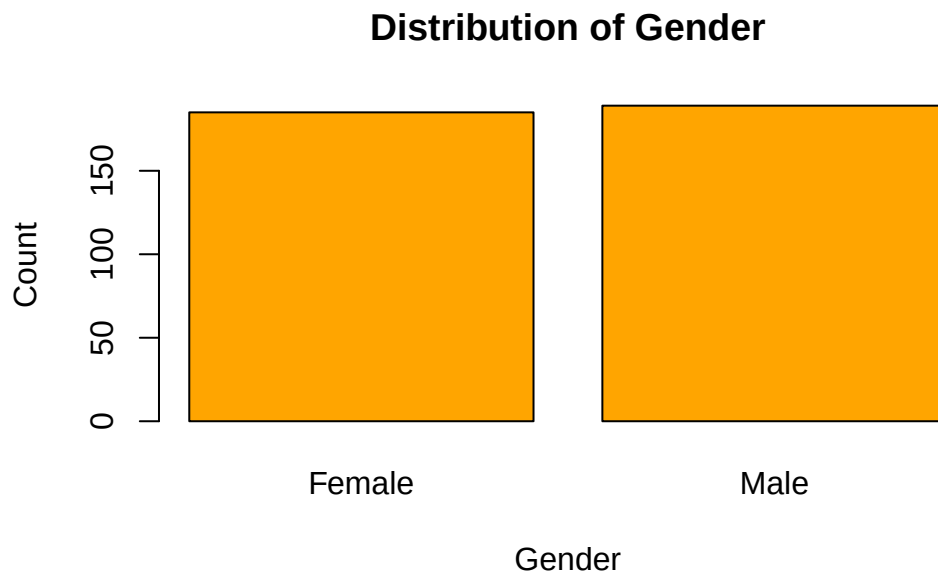
This dataset includes the information of individuals who are between 27 and 59 years old. In this case, based on the type of occupations, the dataset only provides information of people already in the workforce, which does not include students and older people who may experience different sleep or health conditions. Additionally, those who are between 42 and 45 years old are among the top participants in this dataset, with fewer people in their late 20s.

Gender Distribution of Study Participants

```
z <- sleep_data$Gender
gender_counts <- table(sleep_data$Gender)
gender_counts
```

Female	Male
185	189

```
barplot(gender_counts,
       main = "Distribution of Gender",
       xlab = "Gender",
       ylab = "Count",
       col = "orange")
```



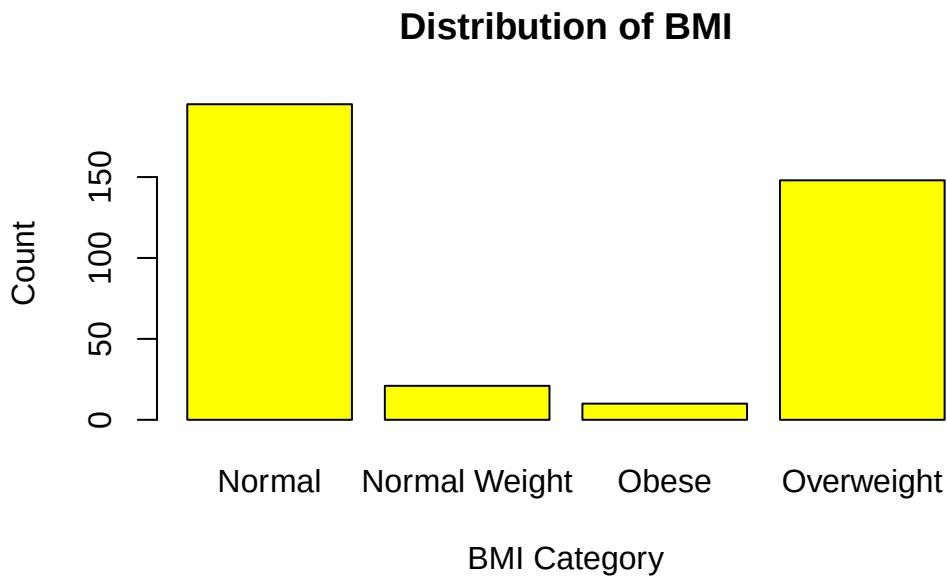
The gender distribution is more balanced compared to the distributions for occupation and age. The dataset includes 185 females, and 189 males. This means that there is fair representation across genders, which helps minimize bias. Thus, this distribution allows for better comparison between females and males.

BMI Distribution among Study Participants

```
j <- sleep_data$`BMI Category`
bmi_counts <- table(sleep_data$`BMI Category`)
bmi_counts
```

Normal	Normal Weight	Obese	Overweight
195	21	10	148

```
barplot(bmi_counts,
        main = "Distribution of BMI",
        xlab = "BMI Category",
        ylab = "Count",
        col = "yellow")
```



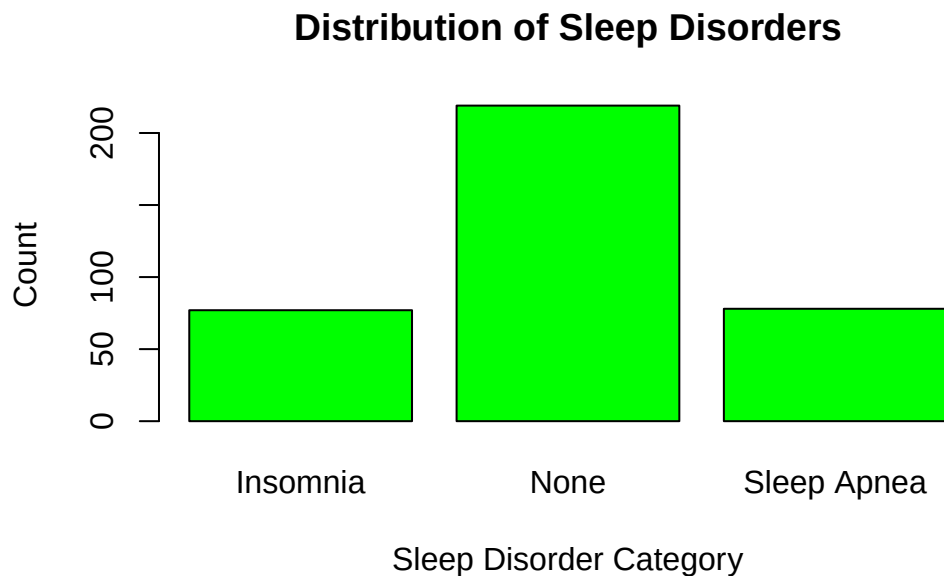
As observed above, the majority of the participants fall between “normal” and “overweight” while only a few reported “normal weight” or being “obese.” All categories are important for this analysis as results for our research questions may vary depending on the individual’s BMI. **Normal weight and normal are possibly the same category which we will feature engineer later*

Sleep Disorder Distribution among Study Participants

```
k <- sleep_data$`Sleep Disorder`
sleep_disorder_counts <- table(sleep_data$`Sleep Disorder`)
sleep_disorder_counts
```

Insomnia	None	Sleep Apnea
77	219	78

```
barplot(sleep_disorder_counts,
       main = "Distribution of Sleep Disorders",
       xlab = "Sleep Disorder Category",
       ylab = "Count",
       col = "green")
```



Sleep disorders are important to understanding how certain individuals function. However, we can observe that this dataset is limited to only two types of disorders: insomnia and sleep apnea. Both seem equally distributed, meaning that around the same number of participants reported having serious sleep conditions. At the same time, we can see that over 200 individuals do not have sleep disorders, which means that more than 50 percent of the participants do not struggle with sleep.

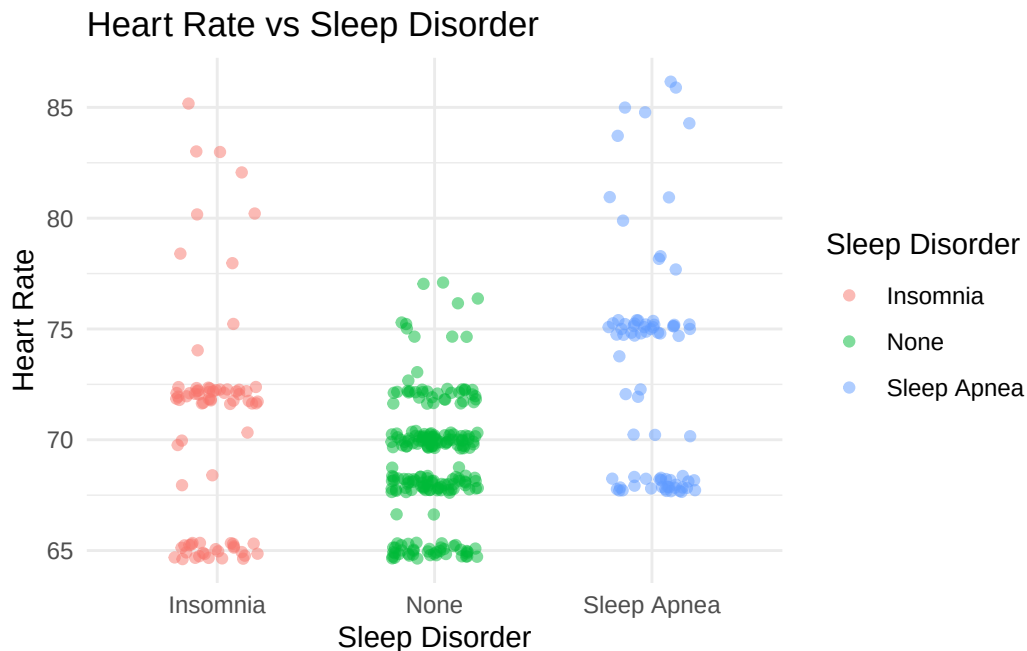
Resting Heart Rate versus Sleep Disorder Status

```
library(ggplot2)

ggplot(sleep_data, aes(x = `Sleep Disorder`, y = `Heart Rate`, color = `Sleep Disorder`)) +
  #spreads points so they don't overlap
  geom_jitter(width = 0.2, alpha = 0.5) +
  labs(title = "Heart Rate vs Sleep Disorder",
```



```
x = "Sleep Disorder",
y = "Heart Rate") +
theme_minimal()
```



Overall, we see that those without a sleep disorder have a lower resting heart rate compared to those who do have a sleeping disorder. Thus, sleep disorders may be correlated with cardiovascular health, specifically heart rate.

Multiple Linear Regression Model: Resting Heart Rate versus Predictor Variables

```
#The following variables are categorical
sleep_data$`BMI Category` <- as.factor(sleep_data$`BMI Category`)
sleep_data$Occupation <- as.factor(sleep_data$Occupation)
sleep_data$`Sleep Disorder` <- as.factor(sleep_data$`Sleep Disorder`)

#blood pressure is not necessarily numeric on its own
#so we need to split it into systolic and diastolic readings
bp_split <- strsplit(as.character(sleep_data$`Blood Pressure`), "/")
sleep_data$Systolic <- as.numeric(sapply(bp_split, `[`, 1))
sleep_data$Diastolic <- as.numeric(sapply(bp_split, `[`, 2))
```

```
#linear regression
#heart rate versus predictor variables
model <- lm(`Heart Rate` ~ `BMI Category` + `Sleep Duration` + Occupation +
            Systolic + Diastolic + `Sleep Disorder` +
            `Quality of Sleep` + `Physical Activity Level` +
            `Stress Level` + `Daily Steps` + Age,
            data = sleep_data)

summary(model)
```

Call:

```
lm(formula = `Heart Rate` ~ `BMI Category` + `Sleep Duration` +
    Occupation + Systolic + Diastolic + `Sleep Disorder` + `Quality of Sleep` +
    `Physical Activity Level` + `Stress Level` + `Daily Steps` +
    Age, data = sleep_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.6309	-0.3238	-0.0927	0.1459	5.5609

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	74.2197261	4.5063765	16.470	< 2e-16 ***
`BMI Category`Normal Weight	0.9941701	0.3462093	2.872	0.004333 **
`BMI Category`Obese	13.5977335	0.7084172	19.195	< 2e-16 ***
`BMI Category`Overweight	1.4650445	0.5012160	2.923	0.003692 **
`Sleep Duration`	0.8912050	0.2723086	3.273	0.001171 **
OccupationDoctor	-1.5702366	0.4075064	-3.853	0.000139 ***
OccupationEngineer	-0.2543227	0.4230202	-0.601	0.548091
OccupationLawyer	0.2973631	0.4928133	0.603	0.546634
OccupationManager	3.9635691	1.3264328	2.988	0.003005 **
OccupationNurse	1.6430545	0.5743284	2.861	0.004479 **
OccupationSales Representative	-5.1820148	1.2256423	-4.228	3.01e-05 ***
OccupationSalesperson	-1.0281897	0.4488541	-2.291	0.022575 *
OccupationScientist	3.8856122	0.8471537	4.587	6.28e-06 ***
OccupationSoftware Engineer	0.2639472	0.7658550	0.345	0.730569
OccupationTeacher	-1.6221621	0.4545270	-3.569	0.000408 ***
Systolic	0.0232388	0.0890112	0.261	0.794187
Diastolic	-0.1652707	0.1190648	-1.388	0.165998
`Sleep Disorder`None	0.1633121	0.3121297	0.523	0.601153
`Sleep Disorder`Sleep Apnea	1.3052510	0.3414283	3.823	0.000156 ***

`Quality of Sleep`	-0.8086698	0.2704015	-2.991	0.002981	**
`Physical Activity Level`	0.0570378	0.0074145	7.693	1.48e-13	***
`Stress Level`	1.4897677	0.1556592	9.571	< 2e-16	***
`Daily Steps`	-0.0006803	0.0001122	-6.064	3.44e-09	***
Age	-0.0332649	0.0254005	-1.310	0.191185	

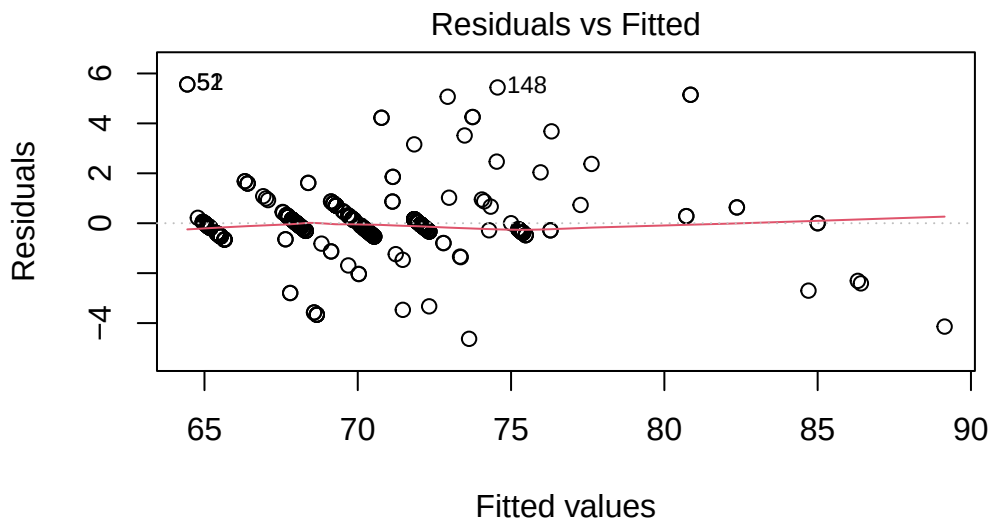
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.261 on 350 degrees of freedom

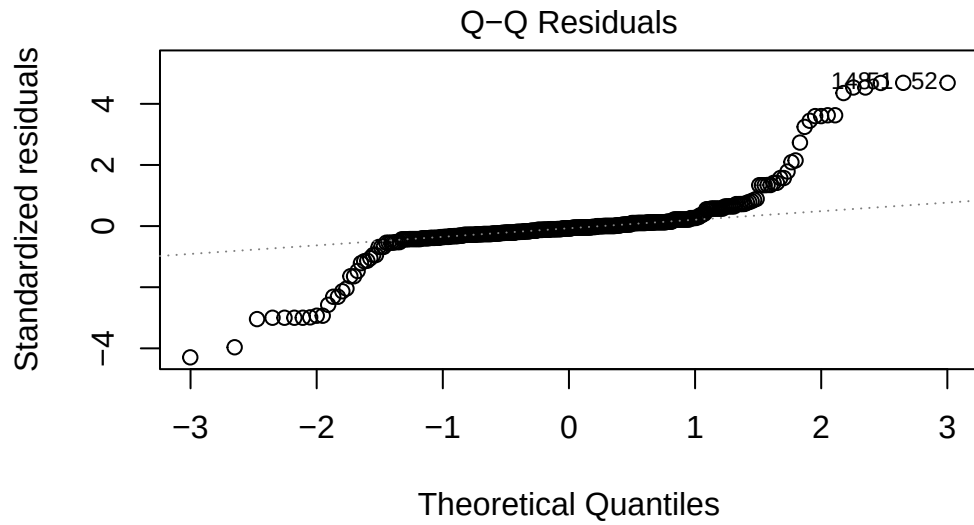
Multiple R-squared: 0.9127, Adjusted R-squared: 0.907

F-statistic: 159.1 on 23 and 350 DF, p-value: < 2.2e-16

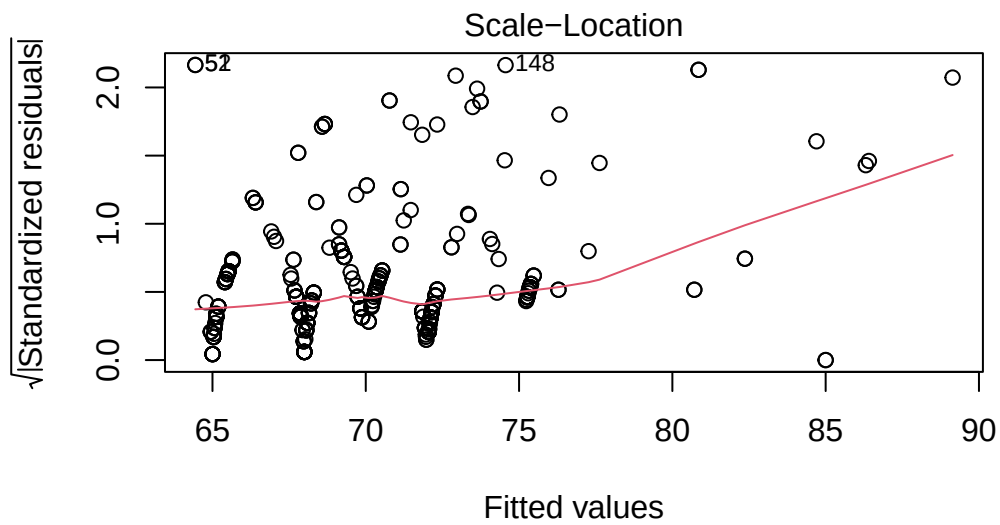
```
plot(model)
```



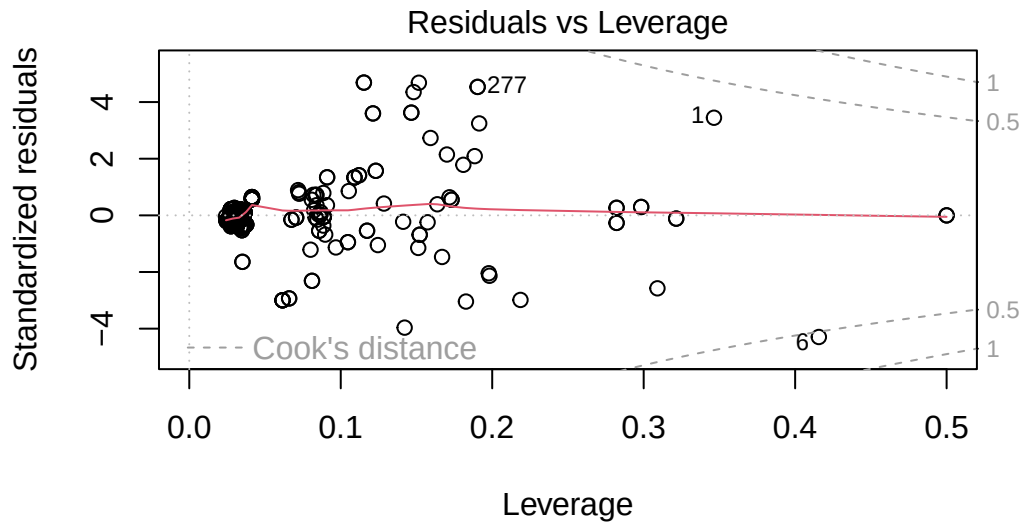
lm(`Heart Rate` ~ `BMI Category` + `Sleep Duration` + Occupation + Systol



lm('Heart Rate' ~ 'BMI Category' + 'Sleep Duration' + Occupation + Systol



lm('Heart Rate' ~ 'BMI Category' + 'Sleep Duration' + Occupation + Systol



Hypotheses

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_j = 0$$

-none of the predictors influence heart rate

$$H_1 : \text{at least one } \beta_j \neq 0$$

-at least one of the predictors does influence heart rate

Fit

Our adjusted R-squared value is 0.907 and it suggests our predictors explain a lot of the variation in our model.

F-statistic

Our f-statistic is large given that it's 159.1 with 23 and 350 degrees of freedom. 23 comes from the number of predictors used in our model and 350 is obtained using $n-p-1$. Here, $n = 374$ and $p = 23$. The f-statistic being large is good because our f-statistic is a ratio of 2 variances (between groups and within groups). The within groups variance is bad because its unexplained variance. Also, our p-value is $< 2.2 \cdot 10^{-16}$ which suggests our model has high statistical significance under an established significance level of $\alpha = 0.05$.

Predictor Variables' Slopes Estimates

Belonging to the obese bmi category has the greatest influence on resting heart rate given that its slope estimate is approximately 13.51. It's also interesting to note that certain occupations do have a significant impact on resting heart rate. Within the occupations, being a scientist and manager increased resting heart rate the most whereas being a sales representative decreased it the most.

Residuals versus Fitted Plot

It shows some random scattering which is good, but oddly the majority is concentrated on the left which we should investigate. Also, upon closer look, there is a slight curve pattern starting at the middle which then dips down. This might suggest some nonlinearity which we should also look into.

QQ Plot

Our QQ plot looks good in the middle given that the points follow the diagonal line. However, on the edges the points deviate greatly which is concerning. The points as a whole follow an S-shape so this suggests having some outliers. We will investigate why this happened and try to implement transformations to get those outlier points to follow the diagonal line closer.

Lasso

Lasso for overall model that doesn't distinguish by sleep disorder.

```
library(glmnet)
library(caret)

#converting categorical variables to factors
sleep_data$`BMI Category` <- as.factor(sleep_data$`BMI Category`)
sleep_data$Occupation <- as.factor(sleep_data$Occupation)
sleep_data$`Sleep Disorder` <- as.factor(sleep_data$`Sleep Disorder`)

#conversion from before
bp_split <- strsplit(as.character(sleep_data$`Blood Pressure`), "/")
sleep_data$Systolic <- as.numeric(sapply(bp_split, `[`, 1))
sleep_data$Diastolic <- as.numeric(sapply(bp_split, `[`, 2))

#predictor matrix
x <- model.matrix(
  `Heart Rate` ~
```

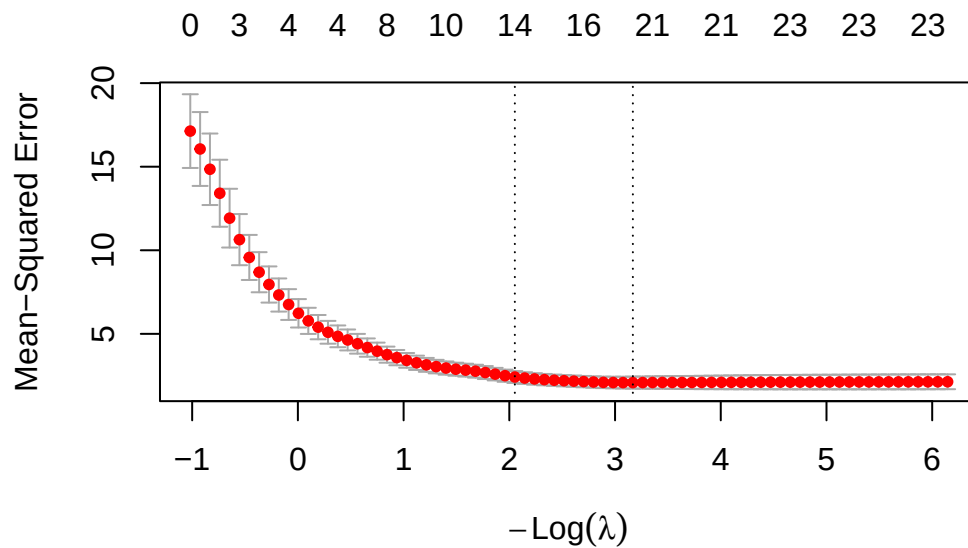
```

  `BMI Category` +
  `Sleep Duration` +
  Occupation +
  Systolic +
  Diastolic +
  `Sleep Disorder` +
  `Quality of Sleep` +
  `Physical Activity Level` +
  `Stress Level` +
  `Daily Steps` +
  Age,
  data = sleep_data
)[-1]
#response
y <- sleep_data$`Heart Rate`

#cv lasso
set.seed(123)
cv_lasso <- cv.glmnet(
  x,
  y,
  alpha = 1,      # alpha = 1 means LASSO
  nfolds = 10
)

#cv error
plot(cv_lasso)

```



```
#lambda with lowest cv error
cv_lasso$lambda.min
```

```
[1] 0.04206058
```

```
#final lasso model
lasso_model <- glmnet(
  x,
  y,
  alpha = 1,
  lambda = cv_lasso$lambda.min
)

coef(lasso_model)
```

```
24 x 1 sparse Matrix of class "dgCMatrix"
                                     s0
(Intercept)                        72.5153241636
`BMI Category`Normal Weight         1.0447700718
`BMI Category`Obese                 12.3053717485
`BMI Category`Overweight             .
`Sleep Duration`                     .
```


OccupationDoctor	-0.9160630343
OccupationEngineer	-0.0388171904
OccupationLawyer	.
OccupationManager	4.2103403567
OccupationNurse	1.1047244554
OccupationSales Representative	-3.2902354338
OccupationSalesperson	.
OccupationScientist	4.5490755493
OccupationSoftware Engineer	0.4931507048
OccupationTeacher	-1.4912968903
Systolic	-0.0331050834
Diastolic	.
`Sleep Disorder`None	.
`Sleep Disorder`Sleep Apnea	1.2401556501
`Quality of Sleep`	-0.5810656320
`Physical Activity Level`	0.0547287773
`Stress Level`	1.2693012292
`Daily Steps`	-0.0006202292
Age	-0.0061567749

We also did lasso divided by sleep disorder. This is important because we suspect that sleep apnea and insomnia have different predictors.

```
#divide data by status
apnea_data <- subset(
  sleep_data,
  `Sleep Disorder` == "Sleep Apnea"
)
insomnia_data <- subset(
  sleep_data,
  `Sleep Disorder` == "Insomnia"
)

none_data <- subset(
  sleep_data,
  `Sleep Disorder` == "None"
)

#run models on data subsets
run_lasso <- function(data_subset){
  x <- model.matrix(
    `Heart Rate` ~
```

```

    `BMI Category` +
    `Sleep Duration` +
    Occupation +
    Systolic +
    Diastolic +
    `Quality of Sleep` +
    `Physical Activity Level` +
    `Stress Level` +
    `Daily Steps` +
    Age,
    data = data_subset
  )[, -1]

#response
y <- data_subset$`Heart Rate`

#cv
cv_fit <- cv.glmnet(
  x,
  y,
  alpha = 1,
  nfolds = 10
)

#final model, lambda selection for penalty strength
final_model <- glmnet(
  x,
  y,
  alpha = 1,
  lambda = cv_fit$lambda.min
)

return(list(
  cv = cv_fit,
  model = final_model
))
}

#3 models
apnea_lasso <- run_lasso(apnea_data)
insomnia_lasso <- run_lasso(insomnia_data)
none_lasso <- run_lasso(none_data)

```

```
coef(apnea_lasso$model)
```

```
22 x 1 sparse Matrix of class "dgCMatrix"
                                     s0
(Intercept)                25.4941665711
`BMI Category`Normal Weight    8.5490220940
`BMI Category`Obese           11.1442922786
`BMI Category`Overweight      .
`Sleep Duration`              .
OccupationDoctor              1.7379818985
OccupationEngineer            -0.8595449829
OccupationLawyer              .
OccupationManager             .
OccupationNurse               -1.6306767921
OccupationSales Representative -1.8395020430
OccupationSalesperson         0.0122406188
OccupationScientist           8.7553967562
OccupationSoftware Engineer   .
OccupationTeacher             4.3962488514
Systolic                     0.2067949036
Diastolic                     .
`Quality of Sleep`            0.6586291417
`Physical Activity Level`      0.0877412407
`Stress Level`                2.1908390456
`Daily Steps`                 -0.0009633887
Age                           0.0491055301
```

```
coef(insomnia_lasso$model)
```

```
22 x 1 sparse Matrix of class "dgCMatrix"
                                     s0
(Intercept)                90.440220254
`BMI Category`Normal Weight    .
`BMI Category`Obese           9.907348715
`BMI Category`Overweight      1.625511972
`Sleep Duration`              .
OccupationDoctor              -0.782051985
OccupationEngineer            2.199900358
OccupationLawyer              2.242323327
OccupationManager             .
OccupationNurse               3.153304831
```

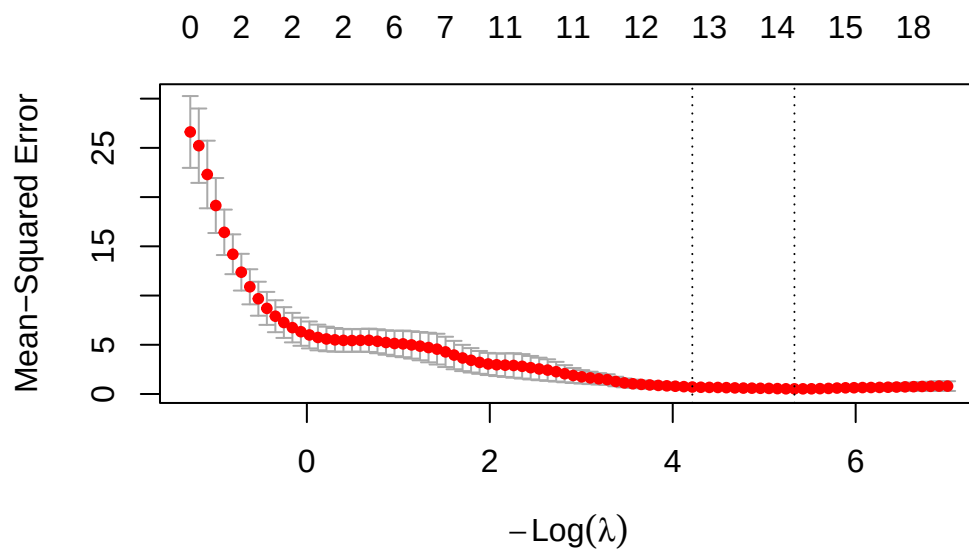
OccupationSales Representative	.
OccupationSalesperson	-1.366009851
OccupationScientist	.
OccupationSoftware Engineer	-3.214568193
OccupationTeacher	-2.961305129
Systolic	-0.098360107
Diastolic	.
`Quality of Sleep`	-1.337473349
`Physical Activity Level`	0.114818754
`Stress Level`	1.188756013
`Daily Steps`	-0.001802694
Age	-0.012143183

```
coef(none_lasso$model)
```

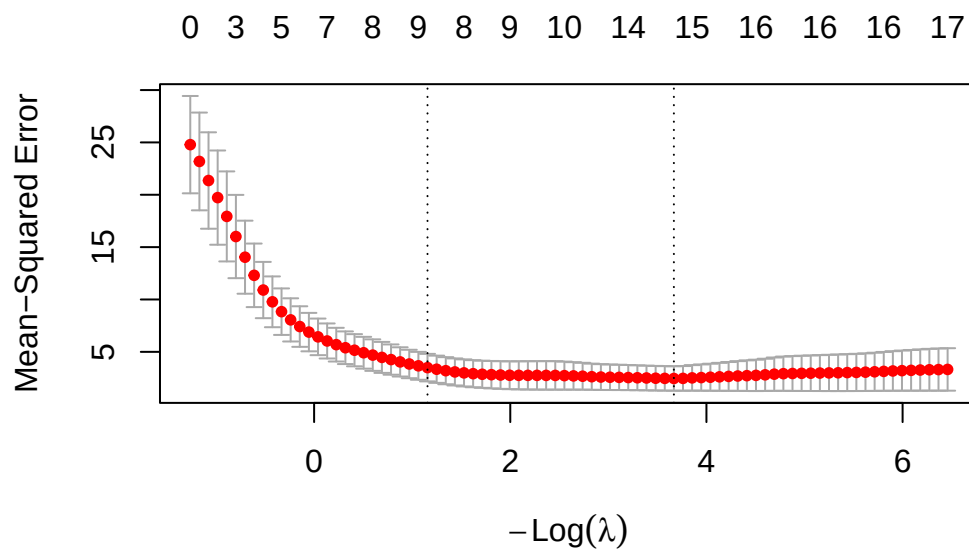
22 x 1 sparse Matrix of class "dgCMatrix"

	s0
(Intercept)	70.77022052
`BMI Category`Normal Weight	0.13066013
`BMI Category`Obese	.
`BMI Category`Overweight	0.86150404
`Sleep Duration`	.
OccupationDoctor	-0.99247433
OccupationEngineer	.
OccupationLawyer	.
OccupationManager	4.36957984
OccupationNurse	1.51679379
OccupationSales Representative	.
OccupationSalesperson	-0.25945324
OccupationScientist	3.98182139
OccupationSoftware Engineer	2.17664300
OccupationTeacher	-0.43107884
Systolic	-0.02308061
Diastolic	-0.01829752
`Quality of Sleep`	-0.63399341
`Physical Activity Level`	0.02653807
`Stress Level`	1.16669332
`Daily Steps`	.
Age	.

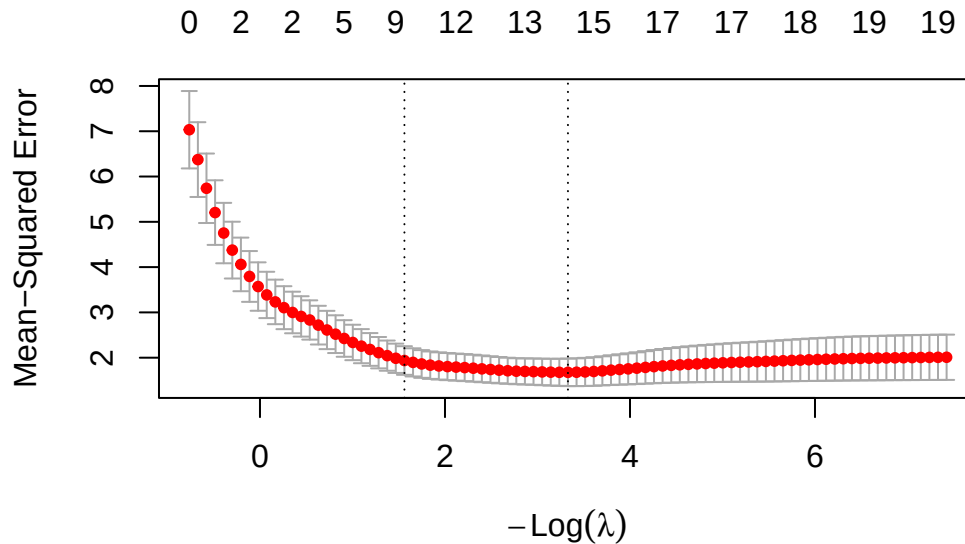
```
plot(apnea_lasso$cv)
```



```
plot(insomnia_lasso$cv)
```



```
plot(none_lasso$cv)
```



Ridge Regression

As observed in the value of the adjusted R-squared (0.907), the model is explaining a lot of the variability in the model. However, the value is closer to 1, which makes it suspiciously high considering that the data only includes about 374 observations and 23 predictors. In this case, many of the predictors may be correlated with each other, resulting in multicollinearity. Additionally, in the analysis section we can observe that the residual vs fitted plot displays some curvature, which violates the linearity assumption. Thus, to address multicollinearity we include a ridge regression.

```
library(glmnet)
x_var <- model.matrix(`Heart Rate` ~ `BMI Category` + `Sleep Duration`
                      + Occupation + Systolic + Diastolic + `Sleep Disorder` +
                      `Quality of Sleep` + `Physical Activity Level` +
                      `Stress Level` + `Daily Steps` + Age,
                      data = sleep_data)[, -1]
y_var <- sleep_data$`Heart Rate`
lambda_seq <- 10^seq(2, -2, by = -.1)

ridge_cv <- cv.glmnet(x_var, y_var, alpha = 0, lambda = lambda_seq)
```

```
best_lambda <- ridge_cv$lambda.min
cat("Best lambda:", best_lambda)
```

Best lambda: 0.1

```
best_ridge <- glmnet(x_var, y_var, alpha = 0, lambda = best_lambda)

coef(best_ridge)
```

24 x 1 sparse Matrix of class "dgCMatrix"

	s0
(Intercept)	76.6383381018
`BMI Category`Normal Weight	1.0758219749
`BMI Category`Obese	12.7267358346
`BMI Category`Overweight	0.8712762680
`Sleep Duration`	0.4408470282
OccupationDoctor	-1.2174951704
OccupationEngineer	-0.1648899397
OccupationLawyer	0.2910025405
OccupationManager	4.1153934345
OccupationNurse	1.2485632801
OccupationSales Representative	-4.6230038965
OccupationSalesperson	-0.7026816950
OccupationScientist	3.7976090184
OccupationSoftware Engineer	0.4308520828
OccupationTeacher	-1.9429185010
Systolic	-0.0194527101
Diastolic	-0.0634309520
`Sleep Disorder`None	0.1281232105
`Sleep Disorder`Sleep Apnea	1.4089256922
`Quality of Sleep`	-0.9519669853
`Physical Activity Level`	0.0534537289
`Stress Level`	1.1901346104
`Daily Steps`	-0.0006307881
Age	-0.0226659759

```
pred <- predict(best_ridge, s = best_lambda, newx = x_var)

rss <- sum((pred - y_var)^2)
tss <- sum((y_var - mean(y_var))^2)
```

```
rsq <- 1 - rss/tss
cat("Ridge R2:", round(rsq, 4))
```

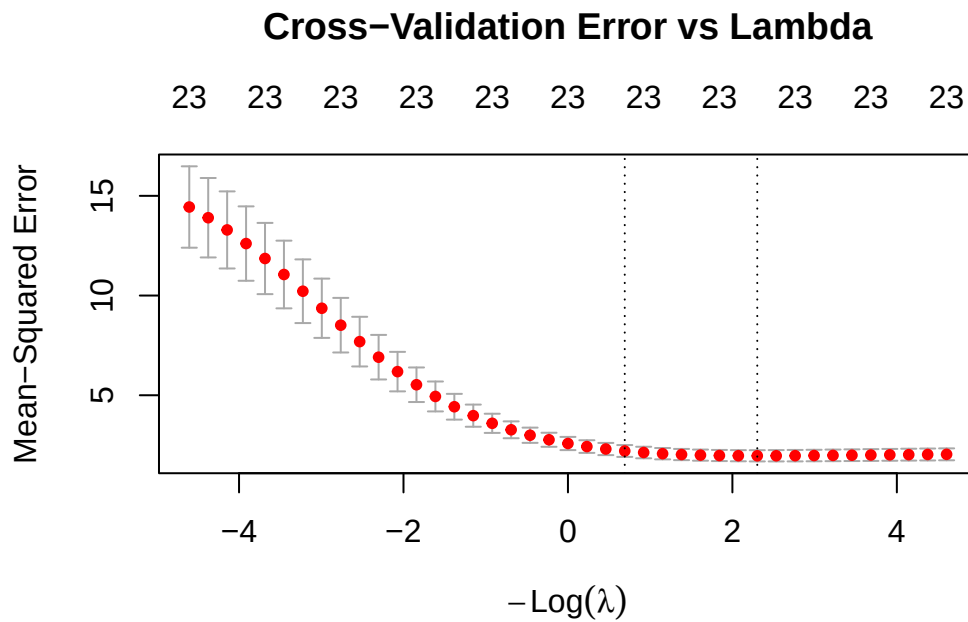
Ridge R²: 0.91

```
summary(x_var)
```

`BMI Category`Normal Weight		`BMI Category`Obese		`BMI Category`Overweight	
Min.	:0.00000	Min.	:0.00000	Min.	:0.0000
1st Qu.:	0.00000	1st Qu.:	0.00000	1st Qu.:	0.0000
Median	:0.00000	Median	:0.00000	Median	:0.0000
Mean	:0.05615	Mean	:0.02674	Mean	:0.3957
3rd Qu.:	0.00000	3rd Qu.:	0.00000	3rd Qu.:	1.0000
Max.	:1.00000	Max.	:1.00000	Max.	:1.0000
`Sleep Duration` OccupationDoctor OccupationEngineer OccupationLawyer					
Min.	:5.800	Min.	:0.0000	Min.	:0.0000
1st Qu.:	6.400	1st Qu.:	0.0000	1st Qu.:	0.0000
Median	:7.200	Median	:0.0000	Median	:0.0000
Mean	:7.132	Mean	:0.1898	Mean	:0.1684
3rd Qu.:	7.800	3rd Qu.:	0.0000	3rd Qu.:	0.0000
Max.	:8.500	Max.	:1.0000	Max.	:1.0000
OccupationManager OccupationNurse OccupationSales Representative					
Min.	:0.000000	Min.	:0.0000	Min.	:0.000000
1st Qu.:	0.000000	1st Qu.:	0.0000	1st Qu.:	0.000000
Median	:0.000000	Median	:0.0000	Median	:0.000000
Mean	:0.002674	Mean	:0.1952	Mean	:0.005348
3rd Qu.:	0.000000	3rd Qu.:	0.0000	3rd Qu.:	0.000000
Max.	:1.000000	Max.	:1.0000	Max.	:1.000000
OccupationSalesperson OccupationScientist OccupationSoftware Engineer					
Min.	:0.00000	Min.	:0.0000	Min.	:0.0000
1st Qu.:	0.00000	1st Qu.:	0.0000	1st Qu.:	0.0000
Median	:0.00000	Median	:0.0000	Median	:0.0000
Mean	:0.08556	Mean	:0.0107	Mean	:0.0107
3rd Qu.:	0.00000	3rd Qu.:	0.0000	3rd Qu.:	0.0000
Max.	:1.00000	Max.	:1.0000	Max.	:1.0000
OccupationTeacher Systolic Diastolic `Sleep Disorder`None					
Min.	:0.000	Min.	:115.0	Min.	:75.00
1st Qu.:	0.000	1st Qu.:	125.0	1st Qu.:	80.00
Median	:0.000	Median	:130.0	Median	:85.00
Mean	:0.107	Mean	:128.6	Mean	:84.65
3rd Qu.:	0.000	3rd Qu.:	135.0	3rd Qu.:	90.00
				Mean	:0.5856
				3rd Qu.:	1.0000

Max. :1.000	Max. :142.0	Max. :95.00	Max. :1.0000
`Sleep Disorder`	`Sleep Apnea`	`Quality of Sleep`	`Physical Activity Level`
Min. :0.0000	Min. :4.000	Min. :30.00	
1st Qu.:0.0000	1st Qu.:6.000	1st Qu.:45.00	
Median :0.0000	Median :7.000	Median :60.00	
Mean :0.2086	Mean :7.313	Mean :59.17	
3rd Qu.:0.0000	3rd Qu.:8.000	3rd Qu.:75.00	
Max. :1.0000	Max. :9.000	Max. :90.00	
`Stress Level`	`Daily Steps`	Age	
Min. :3.000	Min. : 3000	Min. :27.00	
1st Qu.:4.000	1st Qu.: 5600	1st Qu.:35.25	
Median :5.000	Median : 7000	Median :43.00	
Mean :5.385	Mean : 6817	Mean :42.18	
3rd Qu.:7.000	3rd Qu.: 8000	3rd Qu.:50.00	
Max. :8.000	Max. :10000	Max. :59.00	

```
plot(ridge_cv)
title("Cross-Validation Error vs Lambda", line = 3)
```



Smoothing

```
library(splines)

# fit spline model - bs() means "basis spline"
spline_model <- lm(`Heart Rate` ~
  bs(`Physical Activity Level`, df = 4) +
  bs(`Stress Level`, df = 4) +
  bs(`Sleep Duration`, df = 4) +
  `BMI Category` +
  Occupation +
  `Sleep Disorder` +
  Age,
  data = sleep_data)

summary(spline_model)
```

Call:

```
lm(formula = `Heart Rate` ~ bs(`Physical Activity Level`, df = 4) +
  bs(`Stress Level`, df = 4) + bs(`Sleep Duration`, df = 4) +
  `BMI Category` + Occupation + `Sleep Disorder` + Age, data = sleep_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-5.7468	-0.4576	-0.0790	0.4059	6.2844

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	66.40309	1.74346	38.087	< 2e-
16 ***				
bs(`Physical Activity Level`, df = 4)1	5.81178	1.29465	4.489	9.76e-
06 ***				
bs(`Physical Activity Level`, df = 4)2	-2.26572	1.53673	-1.474	0.14129
bs(`Physical Activity Level`, df = 4)3	-1.46904	1.38485	-1.061	0.28953
bs(`Physical Activity Level`, df = 4)4	1.26566	0.43357	2.919	0.00374 **
bs(`Stress Level`, df = 4)1	1.18508	0.84950	1.395	0.16391
bs(`Stress Level`, df = 4)2	-0.65684	1.32253	-0.497	0.61975
bs(`Stress Level`, df = 4)3	8.79273	1.07160	8.205	4.62e-
15 ***				
bs(`Stress Level`, df = 4)4	7.54183	1.06721	7.067	8.82e-
12 ***				

```

bs(`Sleep Duration`, df = 4)1      -0.27148      1.42341     -0.191      0.84885
bs(`Sleep Duration`, df = 4)2        4.10509      1.66309      2.468      0.01406 *
bs(`Sleep Duration`, df = 4)3        3.79136      1.52740      2.482      0.01353 *
bs(`Sleep Duration`, df = 4)4        2.69875      1.46832      1.838      0.06692 .
`BMI Category`Normal Weight      -0.05436      0.37666     -0.144      0.88532
`BMI Category`Obese      15.52543      0.53874     28.818     < 2e-
16 ***
`BMI Category`Overweight        0.18577      0.59761      0.311      0.75610
OccupationDoctor      -0.62404      0.50223     -1.243      0.21488
OccupationEngineer     -0.42312      0.36445     -1.161      0.24645
OccupationLawyer       -0.31204      0.42307     -0.738      0.46129
OccupationManager       6.74339      1.35221      4.987     9.74e-
07 ***
OccupationNurse        2.47340      0.48318      5.119     5.11e-
07 ***
OccupationSales Representative  -3.77334      1.18577     -3.182      0.00159 **
OccupationSalesperson   -1.46448      0.45936     -3.188      0.00156 **
OccupationScientist      6.01734      0.94951      6.337     7.31e-
10 ***
OccupationSoftware Engineer    1.11625      0.85075      1.312      0.19036
OccupationTeacher       -2.07339      0.42115     -4.923     1.32e-
06 ***
`Sleep Disorder`None        0.79129      0.29665      2.667      0.00800 **
`Sleep Disorder`Sleep Apnea    1.72250      0.35727      4.821     2.14e-
06 ***
Age      -0.08567      0.02774     -3.088      0.00218 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Residual standard error: 1.23 on 345 degrees of freedom
Multiple R-squared: 0.9182, Adjusted R-squared: 0.9115
F-statistic: 138.2 on 28 and 345 DF, p-value: < 2.2e-16

Physical Activity

```

activity_range <- seq(min(sleep_data$`Physical Activity Level`),
                      max(sleep_data$`Physical Activity Level`),
                      length.out = 100)

pred_data <- data.frame(
  `Physical Activity Level` = activity_range,
  `Stress Level` = mean(sleep_data$`Stress Level`),

```

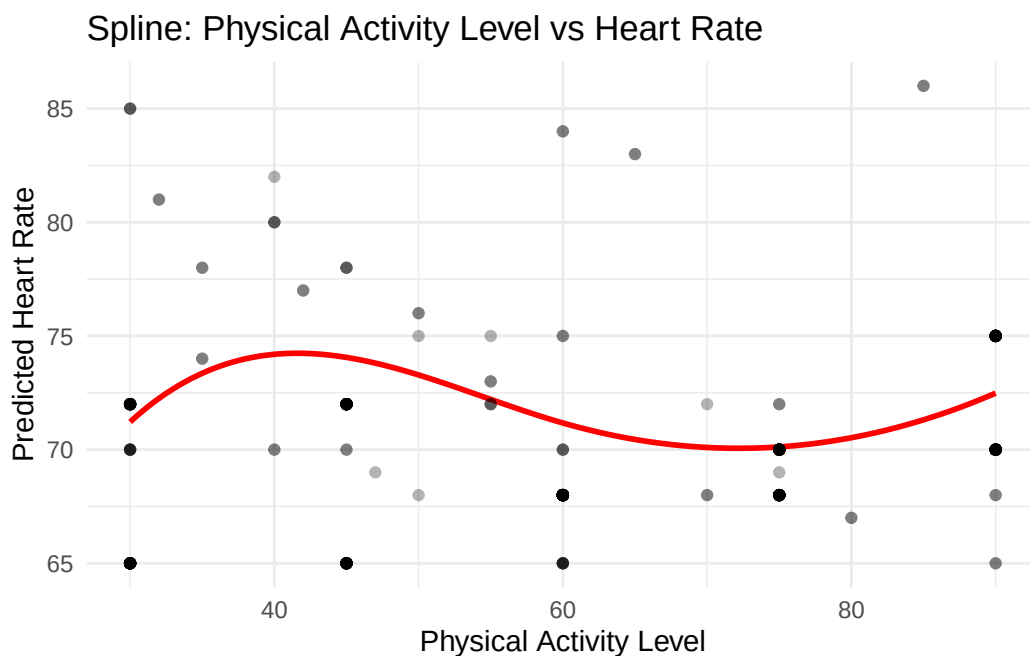
```

`Sleep Duration` = mean(sleep_data$`Sleep Duration`),
`BMI Category` = "Normal",
Occupation = "Nurse",
`Sleep Disorder` = "None",
Age = mean(sleep_data$Age),
check.names = FALSE
)

pred_data$predicted <- predict(spline_model, newdata = pred_data)

ggplot(pred_data, aes(x = `Physical Activity Level`, y = predicted)) +
  geom_line(color = "red", linewidth = 1) +
  geom_point(data = sleep_data,
            aes(x = `Physical Activity Level`, y = `Heart Rate`),
            alpha = 0.3, color = "black") +
  labs(title = "Spline: Physical Activity Level vs Heart Rate",
       x = "Physical Activity Level",
       y = "Predicted Heart Rate") +
  theme_minimal()

```



Stress level

```

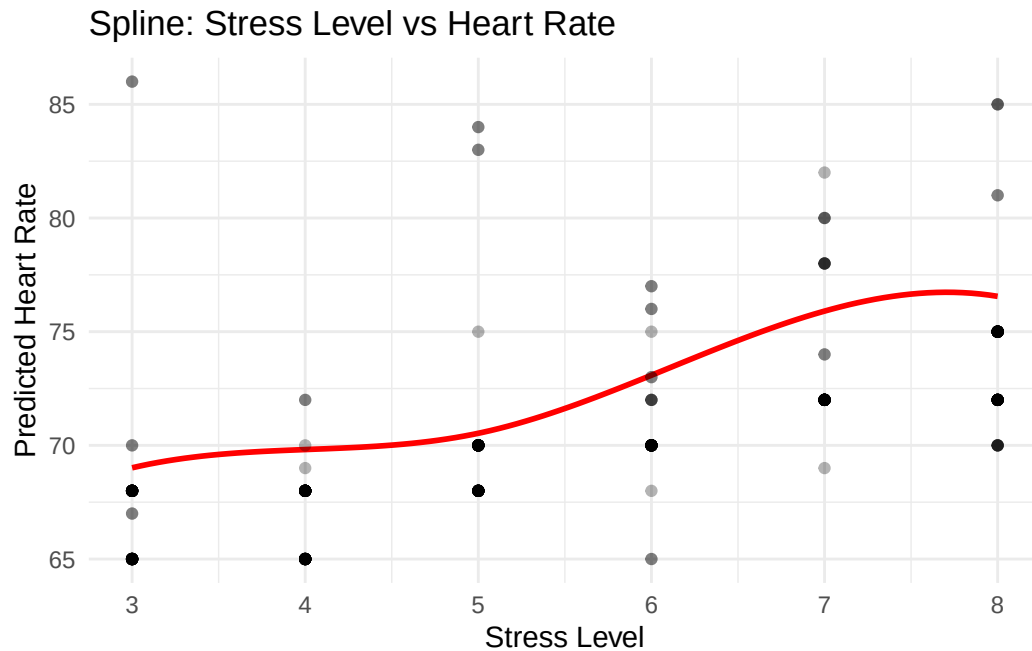
stress_range <- seq(min(sleep_data$`Stress Level`),
                    max(sleep_data$`Stress Level`),
                    length.out = 100)

pred_data2 <- data.frame(
  `Physical Activity Level` = mean(sleep_data$`Physical Activity Level`),
  `Stress Level` = stress_range,
  `Sleep Duration` = mean(sleep_data$`Sleep Duration`),
  `BMI Category` = "Normal",
  Occupation = "Nurse",
  `Sleep Disorder` = "None",
  Age = mean(sleep_data$Age),
  check.names = FALSE
)

pred_data2$predicted <- predict(spline_model, newdata = pred_data2)

ggplot(pred_data2, aes(x = `Stress Level`, y = predicted)) +
  geom_line(color = "red", linewidth = 1) +
  geom_point(data = sleep_data,
            aes(x = `Stress Level`, y = `Heart Rate`),
            alpha = 0.3, color = "black") +
  labs(title = "Spline: Stress Level vs Heart Rate",
       x = "Stress Level",
       y = "Predicted Heart Rate") +
  theme_minimal()

```



Sleep duration

```
duration_range <- seq(min(sleep_data$`Sleep Duration`),
                      max(sleep_data$`Sleep Duration`),
                      length.out = 100)

pred_data3 <- data.frame(
  `Physical Activity Level` = mean(sleep_data$`Physical Activity Level`),
  `Stress Level` = mean(sleep_data$`Stress Level`),
  `Sleep Duration` = duration_range,
  `BMI Category` = "Normal",
  Occupation = "Nurse",
  `Sleep Disorder` = "None",
  Age = mean(sleep_data$Age),
  check.names = FALSE
)

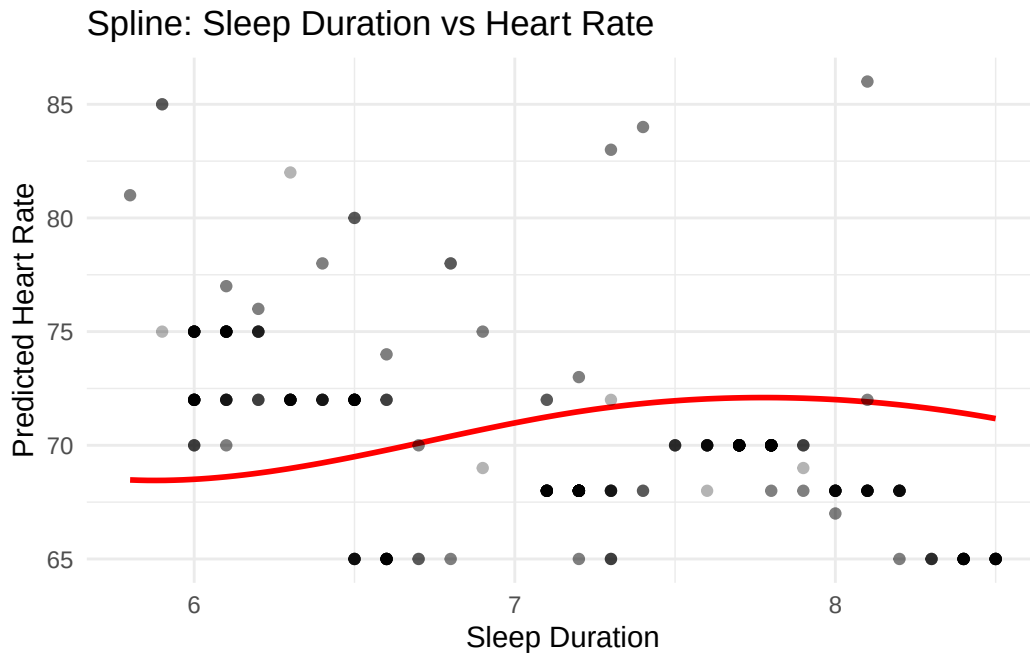
pred_data3$predicted <- predict(spline_model, newdata = pred_data3)

ggplot(pred_data3, aes(x = `Sleep Duration`, y = predicted)) +
  geom_line(color = "red", linewidth = 1) +
  geom_point(data = sleep_data,
            aes(x = `Sleep Duration`, y = `Heart Rate`),
```

```

    alpha = 0.3, color = "black") +
labs(title = "Spline: Sleep Duration vs Heart Rate",
     x = "Sleep Duration",
     y = "Predicted Heart Rate") +
theme_minimal()

```



Multinomial Logistic Regression

```

library(nnet)
#multinomial logistic regression function
#making sure outcome is a factor
sleep_data$`Sleep Disorder` <- factor(sleep_data$`Sleep Disorder`,
                                     levels = c("None", "Insomnia", "Sleep Apnea"))
#fitting our multinomial logistic regression
sleep_disorder_model <- multinom(`Sleep Disorder` ~
    Age + Gender + Occupation +
    `Sleep Duration` + `Quality of Sleep` +
    `Physical Activity Level` + `Stress Level` +
    `BMI Category` + Systolic + Diastolic +
    `Heart Rate` + `Daily Steps`,
    data = sleep_data)

```

```
# weights: 75 (48 variable)
initial value 410.880996
iter 10 value 219.710900
iter 20 value 118.173567
iter 30 value 99.130336
iter 40 value 96.476238
iter 50 value 96.153084
iter 60 value 95.897208
iter 70 value 95.712786
iter 80 value 95.121192
iter 90 value 93.822342
iter 100 value 93.039110
final value 93.039110
stopped after 100 iterations
```

```
summary(sleep_disorder_model)
```

Call:

```
multinom(formula = `Sleep Disorder` ~ Age + Gender + Occupation +
  `Sleep Duration` + `Quality of Sleep` + `Physical Activity Level` +
  `Stress Level` + `BMI Category` + Systolic + Diastolic +
  `Heart Rate` + `Daily Steps`, data = sleep_data)
```

Coefficients:

	(Intercept)	Age	GenderMale	OccupationDoctor
Insomnia	-169.4202	0.16835530	-0.991908	-33.36640
Sleep Apnea	-177.2354	-0.09576782	-1.068213	-17.00251
	OccupationEngineer	OccupationLawyer	OccupationManager	
Insomnia	-18.851643	-20.130805	-24.162729	
Sleep Apnea	-9.516054	-9.703879	-8.683424	
	OccupationNurse	OccupationSales Representative		
Insomnia	-26.582427	-37.08461		
Sleep Apnea	-6.165916	31.51608		
	OccupationSalesperson	OccupationScientist		
Insomnia	-12.788934	-45.865637		
Sleep Apnea	5.256333	-7.859841		
	OccupationSoftware Engineer	OccupationTeacher	`Sleep Duration`	
Insomnia	-34.98234	-14.941311	-1.810161	
Sleep Apnea	-19.24793	3.165768	4.193850	
	`Quality of Sleep`	`Physical Activity Level`	`Stress Level`	
Insomnia	0.4434769	-0.01703747	2.934760	
Sleep Apnea	0.5817862	-0.10011326	2.721276	

	`BMI Category`Normal Weight	`BMI Category`Obese		
Insomnia	-18.20190	23.11547		
Sleep Apnea	-32.06095	23.22554		
	`BMI Category`Overweight	Systolic	Diastolic	`Heart Rate`
Insomnia	-19.71730	-0.009780459	2.312815	0.2760854
Sleep Apnea	-24.76595	-3.116166069	5.894983	1.1118035
	`Daily Steps`			
Insomnia	-0.004627695			
Sleep Apnea	-0.003316315			

Std. Errors:

	(Intercept)	Age	GenderMale	OccupationDoctor
Insomnia	0.002952288	0.06594390	0.08908252	0.015648694
Sleep Apnea	0.003292950	0.08774384	0.07325767	0.006305295
	OccupationEngineer	OccupationLawyer	OccupationManager	
Insomnia	0.02027320	0.0182528	6.270112e-10	
Sleep Apnea	0.01076773	0.0208995	1.050473e-08	
	OccupationNurse	OccupationSales Representative		
Insomnia	0.04723417		NaN	
Sleep Apnea	0.06078151		1.613211e-22	
	OccupationSalesperson	OccupationScientist		
Insomnia	0.03383024	5.185934e-12		
Sleep Apnea	0.02874912	1.096985e-03		
	OccupationSoftware Engineer	OccupationTeacher	`Sleep Duration`	
Insomnia	0.0002238089	0.006954177	0.06661753	
Sleep Apnea	0.0002238098	0.013280770	0.06355166	
	`Quality of Sleep`	`Physical Activity Level`	`Stress Level`	
Insomnia	0.09950512	0.02771427	0.1162715	
Sleep Apnea	0.09635816	0.02501247	0.1287820	
	`BMI Category`Normal Weight	`BMI Category`Obese		
Insomnia	0.02792049	0.002044864		
Sleep Apnea	0.02559912	0.002044864		
	`BMI Category`Overweight	Systolic	Diastolic	`Heart Rate`
Insomnia	0.03526581	0.08610617	0.1059037	0.07344904
Sleep Apnea	0.02174347	0.10674121	0.1272233	0.11902618
	`Daily Steps`			
Insomnia	0.0004274557			
Sleep Apnea	0.0003143605			

Residual Deviance: 186.0782

AIC: 282.0782

```
#p-values
#z-score
z <- summary(sleep_disorder_model)$coefficients / summary(sleep_disorder_model)$standard.error
#2-sided hypothesis testing
#checking is estimate is not 0
p_values <- 2 * (1 - pnorm(abs(z)))
p_values
```

	(Intercept)	Age	GenderMale	OccupationDoctor
Insomnia	0 0.0106797		0	0
Sleep Apnea	0 0.2750759		0	0
	OccupationEngineer	OccupationLawyer	OccupationManager	
Insomnia	0	0	0	
Sleep Apnea	0	0	0	
	OccupationNurse	OccupationSales Representative		
Insomnia	0		NaN	
Sleep Apnea	0		0	
	OccupationSalesperson	OccupationScientist		
Insomnia	0	0		
Sleep Apnea	0	0		
	OccupationSoftware Engineer	OccupationTeacher	`Sleep Duration`	
Insomnia	0	0	0	
Sleep Apnea	0	0	0	
	`Quality of Sleep`	`Physical Activity Level`	`Stress Level`	
Insomnia	8.318244e-06	5.387169e-01	0	
Sleep Apnea	1.562805e-09	6.266763e-05	0	
	`BMI Category`Normal Weight	`BMI Category`Obese		
Insomnia	0	0		
Sleep Apnea	0	0		
	`BMI Category`Overweight	Systolic	Diastolic	`Heart Rate`
Insomnia	0 0.909566	0	0.0001706826	
Sleep Apnea	0 0.000000	0	0.0000000000	
	`Daily Steps`			
Insomnia	0			
Sleep Apnea	0			

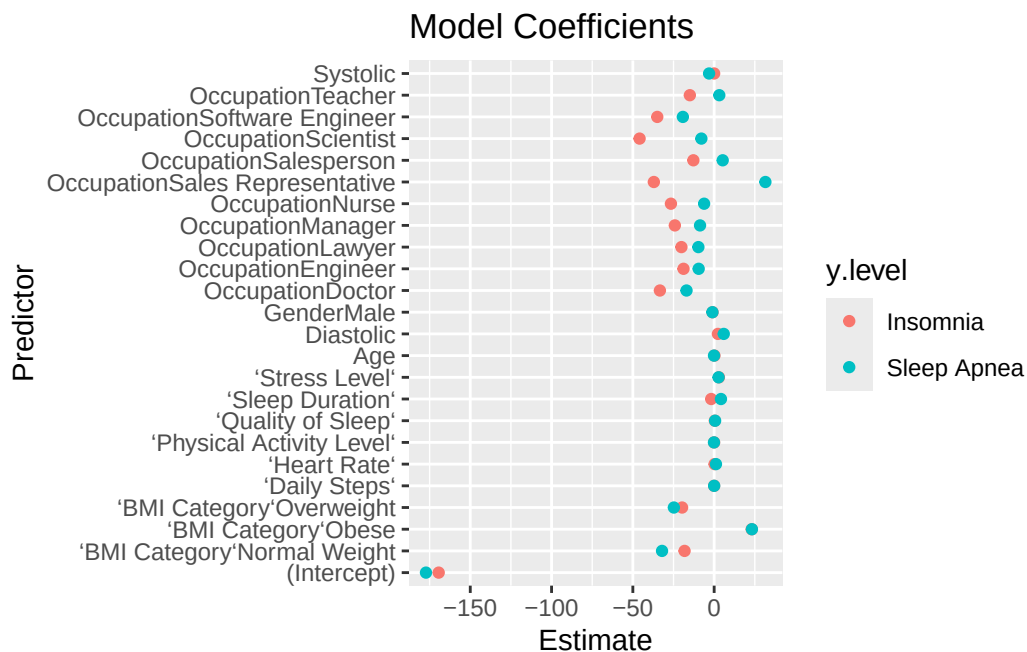
```
library(broom)
library(ggplot2)

#one row for each coefficient
coef_df <- tidy(sleep_disorder_model)
```

```
ggplot
```

```
function (data = NULL, mapping = aes(), ..., environment = parent.frame())  
{  
  UseMethod("ggplot")  
}  
<bytecode: 0x132a5ac50>  
<environment: namespace:ggplot2>
```

```
ggplot(coef_df, aes(x = estimate, y = term, color = y.level)) +  
  geom_point() +  
  labs(title = "Model Coefficients",  
       x = "Estimate",  
       y = "Predictor")
```



Interpretation, Limitations, and Future Directions

The results show that variables such as BMI, stress level, etc. can help explain the variation in resting heart rate. Our model produced a high adjusted R-Squared with a value of 0.907, however, the plots also show violation of the LINE assumptions.

Additionally, there are some limitations in the dataset that may change the results if accounted for. According to the sleep foundation, more than 100 sleeping disorders exist in the medical literature. Yet, our dataset only covers two: sleep apnea and insomnia. While these two sleeping disorders are among the most common, our dataset does not have a wide range of them. This dataset also excludes variables such as income, education levels, blue color jobs, marriage status, number of kids, medical information, ethnic background, and potential genetic predisposition. At this stage, we are unsure if the data reported diagnosed sleep disorders or if the participants self-diagnosed. As we have categorical data, we would need to extend our analysis and implement a multinomial logistic regression model so that it can help with more accurate predictions. Mitigating these limitations may allow for better comparisons and for more accurate interpretations of the response and explanatory variables. Thus, our analysis at this stage should be interpreted carefully.

Conclusion

Our investigation so far shows that resting heart rate is indeed associated with an individual's background information. We came to this conclusion because our model produced a high adjusted R-squared value, 0.907, a large F-statistic, 159 with 23 and 350 degrees of freedom, and an incredibly tiny p-value. Our findings, therefore, suggest a good overall model fit, but given the some of the limitations and violations of the LINE assumptions, they should be interpreted with caution. Despite that, this analysis shows the importance of understanding how an individual's sleeping habits and lifestyle can influence cardiovascular health, in this case, resting heart rate.